5. Project Development Phase

Project Name: FitBuddy - AI Fitness Plan Generator using Gemini Models

Detailed Project Phase Documentation

1. Development Overview

During development, the planned design is converted into a working FitBuddy application. The frontend collects user information and the backend sends a controlled prompt to Gemini.

Tabulation

Module	Function
Frontend	Collect user preferences
Backend	Validate input and call Gemini
AI	Generate plan
Output	Display structured result

Coding / Technical Example

```
from google import genai

client = genai.Client(api_key="YOUR_API_KEY")

response = client.models.generate_content(
    model="gemini-3.8-flash",
    contents=prompt
)

print(response.text)
```

Picture / Diagram

```
FRONTEND → FLASK BACKEND → GEMINI API → RESPONSE → RESULT
```

2. Frontend Development

Create a clean form for experience level, preferred activity, available time, number of days and optional equipment. Provide a Generate Plan button and a readable results area.

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Picture / Diagram

```
FRONTEND → FLASK BACKEND → GEMINI API → RESPONSE → RESULT
```

3. Backend Development

The backend validates input, constructs the prompt, calls the Gemini API and returns the generated result to the frontend.

Tabulation

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AI	Generate plan
Output	Display structured result

Coding / Technical Example

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client = genai.Client(api_key="YOUR_API_KEY")

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Picture / Diagram

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